```
library(DBI)
library(bigrquery)

project_id <- "YOUR_PROJECT"
dataset_id <- "YOUR_DATASET"
billing_project <- "YOUR_BILLING_PROJECT"

con <- dbConnect(
  bigquery(),
  project = project_id,
  dataset = dataset_id,
  billing = billing_project
)



1-

sql_month_keys <- "
CREATE OR REPLACE TABLE
  `YOUR_PROJECT.YOUR_DATASET.design_a_500k_2023_month_keys_25m`

PARTITION BY feature_month
CLUSTER BY party_id

AS

SELECT
    s.party_id,

    DATE(s.observation_date) AS observation_date,

    DATE(s.first_qualifying_alert_date)
       AS first_qualifying_alert_date,

    CAST(s.label_3m AS INT64) AS label_3m,

    month_offset,

    LAST_DAY(
       DATE_SUB(
          DATE_TRUNC(
             DATE(s.observation_date),
             MONTH
          ),
          INTERVAL month_offset MONTH
       ),
       MONTH
```

```
    ) AS feature_month,

    CASE
      WHEN month_offset = 24 THEN 1
      ELSE 0
    END AS seed_month_flag

FROM
    `YOUR_PROJECT.YOUR_DATASET.design_a_500k_2023` s

CROSS JOIN
    UNNEST(GENERATE_ARRAY(0, 24)) AS month_offset
"

DBI::dbExecute(con, sql_month_keys)
```

2-

```
sql_trades_pit <- "
CREATE OR REPLACE TABLE
  `YOUR_PROJECT.YOUR_DATASET.design_a_500k_2023_trades_25m`

PARTITION BY feature_month
CLUSTER BY party_id, acct_id

AS

SELECT

    k.party_id,

    k.observation_date,

    k.first_qualifying_alert_date,

    k.label_3m,

    k.month_offset,

    k.feature_month,

    k.seed_month_flag,

    /* Account identifiers */

    t.trades_seq_num,
```

```
    t.acct_id,
    t.acct_num,
    t.acct_typ_cde,

    /* Account dates */

    t.open_dte,
    t.clsd_dte,
    t.paid_out_dte,
    t.eff_dte,

    /* Main behavioral fields */

    t.acct_mop_cde,

    t.curr_bal_amt,
    t.past_due_amt,
    t.cr_lmt_amt,

    /* Additional useful account-level information */

    t.high_cr_amt,
    t.payt_amt,
    t.payt_due_amt,
    t.payt_dte,

    t.max_delq_amt,
    t.max_delq_dte,
    t.max_delq_rte_cde,

    t.frst_delq_dte,
    t.frst_rpt_dte,

    /* Original source snapshot date */

    DATE(t.data_date) AS source_data_date

FROM
    `YOUR_PROJECT.YOUR_DATASET.design_a_500k_2023_month_keys_25m` k

INNER JOIN
    `YOUR_PROJECT.YOUR_SOURCE_DATASET.trades` t

    ON k.party_id = t.party_id

   AND DATE(t.data_date) = k.feature_month
"
```

```
DBI::dbExecute(con, sql_trades_pit)
```